

TWILIGHT

Terminator Web Interface for Local Inclination Geometry, Heights &
Twilight

Method document

Geometric MVP model for local solar-terminator cross-sections

This document describes the current geometric method used by the TWILIGHT MVP. It is intended as a clear mathematical reference for implementation, review, and validation. The model visualizes local solar-elevation threshold surfaces; it is not a radiative-transfer or sky-brightness model.

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1 Purpose and scope

TWILIGHT is a client-side scientific visualization tool for exploring the local geometry of solar-terminator and twilight-threshold surfaces above a selected ground station.

The current method is designed to answer questions such as:

- where a selected solar-elevation threshold intersects the ground along a local cross-section;
- how the threshold surface rises with altitude;
- what local inclination the profile has at ionospheric reference heights;
- how the geometry changes with station, date, and time.

The tool currently provides a geometric operational visualization. It does not model optical sky brightness, atmospheric scattering, refraction, or radiative transfer.

2 Coordinate and time conventions

All calculations are performed internally in **UTC**. Local station time may be shown for convenience, but UTC remains the computational time base.

Solar position follows the common NOAA-style convention:

- solar elevation e is positive above the local horizon;
- solar azimuth A is measured clockwise from geographic north;
- all angular equations are evaluated in radians unless stated otherwise.

The local coordinate frame is East-North-Up:

- E : east,
- N : north,
- U : up.

The local solar unit vector in ENU coordinates is:

$$\begin{aligned}s_E &= \cos(e) \sin(A), \\ s_N &= \cos(e) \cos(A), \\ s_U &= \sin(e).\end{aligned}$$

The horizontal projection of the solar vector is:

$$\mathbf{p}_{day} = \frac{(s_E, s_N)}{\sqrt{s_E^2 + s_N^2}}$$

This direction points horizontally toward the Sun, that is, toward the daylight side.

3 Cross-section and terminator-front directions

The main cross-section is a local vertical plane:

- perpendicular to the ground;
- perpendicular to the local terminator front;
- aligned with the daylight-night direction.

The current UI convention is:

- negative x : toward daylight;
- positive x : toward night.

If the solar azimuth is A , the direction toward daylight is A . Therefore the positive x -axis of the plotted cross-section, pointing toward night, is:

$$A_{cross} = A + 180^\circ \pmod{360^\circ}$$

The terminator-front direction is perpendicular to the cross-section direction:

$$A_{front} = A_{cross} + 90^\circ \pmod{360^\circ}$$

These azimuths are displayed in the diagnostics and exported data products.

4 Twilight thresholds

The model uses standard solar-elevation threshold definitions.

Threshold	Symbol	Solar elevation
Horizon / geometric terminator	α_H	0°
Civil twilight	α_C	-6°
Nautical twilight	α_N	-12°
Astronomical twilight	α_A	-18°

In the equations below, the selected threshold is denoted by α .

5 Earth model

The MVP uses a spherical Earth approximation:

$$R = 6371 \text{ km}$$

The selected station is treated as a local reference point on the spherical surface. Station altitude is stored and exported, but the current threshold-profile model uses the spherical radius approximation for the cross-section geometry.

This is sufficient for the intended visualization scale, but it is not a full WGS84 ellipsoidal ray-tracing model.

6 Ground displacement approximation

Let:

- x be horizontal distance from the station along the cross-section direction;
- $x > 0$ point toward night;
- e_0 be the solar elevation at the station.

For local ground points displaced by x , the solar elevation is approximated as:

$$e_{ground}(x) = e_0 - \frac{x}{R}$$

where e_0 and x/R are in radians. Moving toward night decreases the apparent solar elevation.

7 Ground intersection of a threshold

At ground level, a threshold line satisfies:

$$e_{ground}(x_g) = \alpha$$

Substituting the ground-displacement approximation gives:

$$\begin{aligned} e_0 - \frac{x_g}{R} &= \alpha, \\ x_g &= R(e_0 - \alpha). \end{aligned}$$

Thus the ground-intersection distance is:

$$x_g = R(e_0 - \alpha)$$

If $x_g = 0$, the station lies directly on that threshold at ground level. If $x_g \neq 0$, the threshold line does not pass through the station at ground level at the selected time, and the tool reports the projected ground-intersection distance along the cross-section.

8 Horizon dip with altitude

At altitude z above a spherical Earth, the geometric dip of the horizon is approximated as:

$$\delta(z) = \arccos\left(\frac{R}{R+z}\right)$$

where R and z use the same length units. In the application, both are in kilometres. For small altitudes:

$$\delta(z) \approx \sqrt{\frac{2z}{R}}$$

This small-altitude behaviour explains the strong curvature of the plotted threshold profiles close to the ground.

9 Threshold curve in the vertical cross-section

At altitude z , the threshold condition used by TWILIGHT is:

$$e_{ground}(x) + \delta(z) = \alpha$$

Substituting the ground-displacement approximation:

$$e_0 - \frac{x}{R} + \delta(z) = \alpha.$$

Solving for x :

$$x(z, \alpha) = R(e_0 + \delta(z) - \alpha)$$

This is the main profile equation used to draw the cross-section curves. Each threshold curve is generated by evaluating $x(z, \alpha)$ for a sampled set of altitudes.

10 Why all threshold curves have the same local slope

The profile equation can be written as:

$$x(z, \alpha) = Re_0 + R\delta(z) - R\alpha$$

For a fixed station and time, e_0 is constant. For a given threshold, $-R\alpha$ is also constant. Changing the threshold from horizon to civil, nautical, or astronomical twilight therefore shifts the curve horizontally without changing its shape.

The local derivative is:

$$\frac{dx}{dz} = R \frac{d\delta}{dz}$$

It does not depend on α . Therefore, in the current spherical geometric model, the local inclination angle of the horizon, civil, nautical, and astronomical threshold profiles is identical at the same altitude. The curves are horizontal translations of one another.

This is a feature of the present geometric model, not a numerical bug. If a later model includes a full 3D shadow volume, ellipsoidal geometry, atmospheric refraction, or ray-tracing effects, the threshold shapes may no longer be simple translations.

11 Profile inclination angle

The profile inclination angle is measured relative to the local horizontal axis in the vertical cross-section.

If the curve is expressed as $x(z)$, then locally:

$$\tan(\theta) = \frac{dz}{dx}$$

and therefore:

$$\theta = \arctan\left(\frac{dz}{dx}\right)$$

Since:

$$\frac{dx}{dz} = R \frac{d\delta}{dz}$$

we can write:

$$\theta(z) = \arctan\left(\frac{1}{R \frac{d\delta}{dz}}\right)$$

For $z > 0$, the derivative of the horizon-dip term is:

$$\frac{d\delta}{dz} = \frac{R}{(R+z)^2 \sqrt{1 - \left(\frac{R}{R+z}\right)^2}}$$

12 Ground-angle interpretation

At the exact ground surface, $z = 0$, the horizon-dip approximation behaves as:

$$\delta(z) \approx \sqrt{\frac{2z}{R}}$$

Thus:

$$\frac{d\delta}{dz} \rightarrow \infty \quad \text{as } z \rightarrow 0^+$$

and:

$$\frac{dx}{dz} \rightarrow \infty$$

which implies:

$$\frac{dz}{dx} \rightarrow 0$$

Therefore, in this model, the threshold curve is tangent to the ground at the ground-intersection point, and the limiting ground angle is:

$$\theta_g = 0^\circ$$

Earlier numerical implementations estimated the near-ground angle by a finite difference between $z = 0$ and a small positive altitude. This produced a small non-zero value. Such a number should be interpreted as an effective near-ground angle over a finite altitude interval, not as the mathematical tangent angle exactly at the ground.

The preferred interpretation in the current method is:

- **ground angle:** exact limiting angle at $z = 0$, equal to 0° ;
- **angle at finite altitude:** local inclination computed at a selected altitude, such as 90 km or 110 km.

13 Numerical angle estimates at selected altitudes

For finite altitudes, such as 90 km or 110 km, the application may estimate the local tangent using finite differences.

For an altitude z_0 , a central finite-difference approximation is:

$$\left. \frac{dx}{dz} \right|_{z_0} \approx \frac{x(z_0 + \Delta z) - x(z_0 - \Delta z)}{2\Delta z}$$

Then:

$$\theta(z_0) \approx \arctan \left(\frac{2\Delta z}{x(z_0 + \Delta z) - x(z_0 - \Delta z)} \right)$$

For $z_0 = 0$, this finite-difference estimate is not the exact ground tangent. The exact limiting ground angle is 0° in the present model.

14 Full-profile and ground-zoom views

The cross-section can be displayed in two complementary modes.

14.1 Full profile

The full-profile view shows the large-scale structure of the threshold curves. Typical ranges are:

$$\begin{aligned} x &\in [-2500, +2500] \text{ km,} \\ z &\in [0, 2000] \text{ km.} \end{aligned}$$

A logarithmic-like vertical display scale may be used:

$$y_{plot} = \log_{10}(z + 1)$$

The physical altitude remains in kilometres; only the visual axis mapping is logarithmic-like. This makes the near-ground and ionospheric regions easier to inspect while still showing the topside region.

14.2 Ground zoom

The ground-zoom view is intended for inspecting the near-ground region close to a station or a ground-intersection point. A practical range is:

$$\begin{aligned} x &\in [-25, +25] \text{ km,} \\ z &\in [0, 25] \text{ km.} \end{aligned}$$

Linear axes and equal visual scaling are preferred in this view, so that local inclinations can be interpreted directly.

15 Ionosphere reference bands

The cross-section view may display approximate ionospheric reference bands.

Region	Approximate altitude range
D region	60–90 km
E region	90–150 km
F1 region	150–250 km
F2 region	250–500 km
Topside ionosphere / plasmasphere	500–2000 km

These are visual reference bands only. Their physical boundaries vary with solar activity, season, latitude, local time, and geophysical conditions.

16 Event-time calculation

For morning and evening modes, the application searches for the UTC time on the selected date when solar elevation at the station crosses the selected event anchor threshold.

The root condition is:

$$f(t) = e(t) - \alpha = 0$$

where $e(t)$ is the solar elevation at time t , and α is the selected event threshold.

The application scans the selected UTC day and applies a root-finding method such as bisection to intervals where a crossing is detected.

The crossing is classified as:

- **morning event:** solar elevation increasing through the threshold;
- **evening event:** solar elevation decreasing through the threshold.

If no crossing is detected, the application reports a diagnostic status instead of failing. Examples include no crossing on selected date, always above threshold, always below threshold, polar day, polar night, or missing astronomical twilight.

17 Solar timeline panel

The solar timeline panel plots solar elevation over the selected UTC day and displays the four reference threshold levels:

Threshold	Elevation
Horizon	0°
Civil	−6°
Nautical	−12°
Astronomical	−18°

The selected time is shown as a vertical marker. Diagnostic values may include solar elevation, solar azimuth, cross-section azimuth, terminator-front azimuth, event time, threshold-specific ground-intersection distances, and inclination angles.

18 Limitations of the MVP model

The current method is intentionally simplified.

1. **Spherical Earth approximation.** The current model uses $R = 6371$ km, not a full WGS84 ellipsoid.
2. **No atmospheric refraction.** Optical refraction near the horizon is not currently included.
3. **No atmospheric scattering or sky brightness.** Civil, nautical, and astronomical thresholds are treated as geometric solar-elevation surfaces.
4. **No full 3D ray tracing.** The cross-section is a local geometric approximation, not a global volumetric shadow model.
5. **Threshold curves are parallel translations.** In the current model, all threshold curves have identical shape and local slope at the same altitude.
6. **Ground angle is a limiting quantity.** The exact ground tangent angle is 0°. Non-zero near-ground angles arise only from finite-difference estimates over non-zero altitude intervals.
7. **Map and Sun–Earth panels are schematic.** Until a more rigorous global geometry overlay is implemented, those panels should be treated as visual aids rather than authoritative geometric outputs.

19 Recommended validation checks

Recommended validation steps include:

1. Compare solar elevation and azimuth against a trusted solar-position calculator for several dates, stations, and times.
2. Compare morning/evening threshold event times against independent sunrise, sunset, and twilight calculators.
3. Test mid-latitude normal cases, summer white-night cases, polar day, and polar night.
4. Confirm that no output displays NaN, Infinity, or silently invalid values.
5. Confirm that ground-intersection distances shift consistently when changing threshold level.

6. Confirm that threshold curves have the same shape and slope in the current model, differing only by horizontal displacement.
7. Confirm that the exact ground angle is reported as 0° in the current geometric model.

20 References and background sources

- Reda, I.; Andreas, A. (2008). *Solar Position Algorithm for Solar Radiation Applications*. National Renewable Energy Laboratory, NREL/TP-560-34302.
- Meeus, J. *Astronomical Algorithms*. Willmann-Bell.
- U.S. Naval Observatory definitions of civil, nautical, and astronomical twilight.
- NOAA solar calculator documentation and solar azimuth/elevation conventions.
- Station coordinate sources for LOFAR PL612 Baldy and IGS LAMA should be verified and updated when authoritative station metadata are available.

21 Implementation note

This document describes the MVP geometric method. If a future version introduces a full 3D Earth-shadow model, WGS84 ellipsoidal geometry, atmospheric refraction, or ray tracing through altitude-dependent atmospheric layers, this document should be updated accordingly.

TWILIGHT presently visualizes local geometric solar-elevation threshold surfaces, not physical optical brightness or radiative transfer.